



Sodium modified Fe-Mn microsphere catalyst for Fischer–Tropsch synthesis of light olefins

Hewei Zhang^a, Haitao Zhang^a, Weixin Qian^a, Xian Wu^a, Hongfang Ma^{a,*}, Qiwen Sun^b, Weiyong Ying^a

^a Engineering Research Center of Large Scale Reactor Engineering and Technology, Ministry of Education, State Key Laboratory of Chemical Engineering, School of Chemical Engineering, East China University of Science and Technology, Shanghai 200237, China

^b State Key Laboratory of Coal Liquefaction and Coal Chemical Technology, Shanghai 201203, China

ARTICLE INFO

Keywords:

Fischer–Tropsch synthesis
Fe₃O₄ microsphere catalyst
Mn and Na promoters
Light olefins

ABSTRACT

Both one-pot solvothermal method and impregnation were adopted to synthesize sodium modified Fe-Mn microsphere catalysts for Fischer–Tropsch synthesis (FTS) of light olefins. The effects of Mn and Na promoters on the microsphere morphology, electronic state and catalytic performance of Fe₃O₄ catalysts were evaluated by Ar adsorption-desorption, X-ray diffraction (XRD), scanning electron microscopy (SEM), high-resolution transmission electron microscopy (HRTEM), energy-dispersive spectroscopy mapping (EDS-mapping), H₂ temperature-programmed reduction (H₂-TPR), CO temperature-programmed desorption (CO-TPD), X-ray photoelectron spectroscopy (XPS) and Mössbauer spectra. SEM, TEM and Ar adsorption-desorption showed that catalysts were in a uniform spherical shape with the size of 300–400 nm, while FeMn microsphere owned a smaller size and a larger surface area. Mn promoter could enhance the chemisorption of CO, and suppressed the hydrogenation of olefins, then lead to higher selectivity of light olefins. Na promoter strengthened the FTS reaction activity due to its basicity and electronic donation. Mn₁₅Na₃₂im showed the highest CO conversion of 97.21 % with the light olefin selectivity of 33.02 %.

1. Introduction

Fischer–Tropsch synthesis (FTS) is getting more and more attention for recent years as an important gas to liquid technology. FTS also provides an alternative and promising way to produce synthetic and synthetic fuels from coal, natural gas and renewable biomass resources. Light olefins (C₂–C₄) play a key role in chemical and petrochemical industries to provide polymers, detergents, plastics, textiles and solvents. Light olefins could be obtained by naphtha cracking or by recovery from fluid catalytic cracking [1,2]. Due to the environmental and economic factors [3], processes are needed currently which could synthesize light olefins from alternative feedstocks. FTS offers a route of producing light olefins directly from syngas, and it is also the only way to get light olefins from syngas directly until now.

Metal-based catalysts such as iron and cobalt have been used in the FTS, and they are demonstrated to be appropriate from an economical point of view. Iron-based catalysts own the advantages of flexible operation conditions, low cost, and low sensitivity to poisons. Iron-based

catalysts provide more possible products including paraffins, olefins and alcohols than cobalt-based and ruthenium-based catalysts [4]. FTS reaction over iron-based catalysts is considered to be a surface-sensitive reaction. It means that the catalytic performance strongly depends on the particle size of the active phase [5]. However, it's still limited by low olefin selectivity and high methane selectivity to get light olefins from syngas, and carbon deposition is also a deciding factor. Iron catalysts prepared by coprecipitation method own uncontrollable shape and dispersion, and it may cause poor stability and product selectivity.

Recently, due to the technological and fundamental scientific importance, the functional nanoparticles have become a particularly important area of research. These magnet nanocrystals such as magnetite (Fe₃O₄) or maghemite (γ-Fe₂O₃) own both controllable size and shape [6,7].

Manganese and sodium are widely used over iron-based catalysts for FTS. These two metal promoters can help to improve the light olefin selectivity, FTS reaction activity and stability of Fe catalysts [8,9]. An appropriate amount addition of manganese can increase the surface

* Corresponding author.

E-mail address: mark@ecust.edu.cn (H. Ma).

<https://doi.org/10.1016/j.cattod.2020.07.040>

Received 7 February 2020; Received in revised form 16 June 2020; Accepted 20 July 2020

Available online 29 July 2020

0920-5861/© 2020 Elsevier B.V. All rights reserved.

Table 1

Amounts of materials used to synthesis catalysts.

Catalysts Materials (g)	Mn15	Mn15Na16 _{im}	Mn15Na32 _{im}	Mn15Na32	Na32
FeCl ₃ ·6H ₂ O (g)	5.4	5.4	5.4	5.4	5.4
EG (mL)	100	100	100	100	100
PVP (g)	2.4	2.4	2.4	2.4	2.4
NH ₄ Ac (g)	10.8	10.8	10.8	10.8	10.8
Mn(Ac) ₂ (g)	0.52	0.52	0.52	0.52	\
Na ₂ CO ₃ (g)	\	0.01	0.02	\	\
NaAc (g)	\	\	\	\	0.053

basicity and encourage carburization of Fe catalyst, then enhance the FTS activity. In addition, Mn could also prevent ethylene and propylene from the secondary hydrogenation, then lead to higher selectivity of light olefins [10]. Mn is known to promote the dispersion of Fe and protect the catalyst from deactivation which caused by carbon deposition [11]. However, the formation of Fe-Mn spinel could cause the strong Fe-Mn interaction. Besides, excess amount addition of manganese may slow down the carburization of iron during FTS reaction. While carburization is considered to be necessary to get a high activity. Thus, sodium as the alkali metal is added to the Fe-Mn catalyst system. It's known that sodium addition lead to lower methane selectivity and a significant increase of olefin-to-paraffin ratio. Sodium has also been proposed as an effective promoter to highly increase both the iron carbonization and water-gas shift (WGS) reaction activity, then lead to a higher FTS activity [12]. Ribeiro et al. [13] suggested that alkali promoters such as sodium decrease the strength of —C—O bond, and cause a higher concentration of dissociated CO. Furthermore, alkali metals could also change the local electron density of active transition metals and create blockage to increase activity.

In order to overcome the low selectivity of light olefins and high methane selectivity in FTS reaction based on the above research, Fe₃O₄ nanocatalysts with spherical shape were applied for FTS. In this study, Mn-prompted Fe₃O₄ catalysts were prepared to produce light olefins from syngas with low methane selectivity. Then Na promoter was introduced on the FeMn catalyst to further improve the activity and product selectivity. As-prepared iron-based catalysts were carried out for FTS to investigate the superiority of microsphere catalysts and the role of Mn and Na promoters over Fe₃O₄ Catalysts.

2. Experiments

2.1. Catalyst preparation

Typical Fe₃O₄ nanospheres with Mn supported were obtained by one-step solvothermal synthesis. First, iron (III) chloride hexahydrate (FeCl₃·6H₂O) was added as iron source to ethylene glycol (EG) together with manganese acetate (Mn(Ac)₂) and ammonium acetate (NH₄Ac). The role of ethylene glycol here was both solvent and reducing agent to partly reduce the Fe³⁺ to Fe²⁺. The solution was dissolved by magnetic stirring. Then PVP was added to control the size of magnet. After stirring and dissolving, the obtained solution was transferred into a 200 mL Teflon-lined autoclave for hydrothermal reaction at 200 °C for 12 h. Then the obtained black precipitate after crystallization was washed 6 times with both deionized water and absolute ethanol and then dried under vacuum at 110 °C for 12 h. Finally, the catalysts were calcined at 400 °C for 5 h in a nitrogen atmosphere to obtain the FeMn microsphere catalyst. Then the Na promoter was further introduced by the impregnation method on FeMn catalyst. Afterwards, FeMn catalyst was impregnated with a certain mass fraction of sodium carbonate (Na₂CO₃) solution, then also dried under vacuum and calcined in nitrogen.

In order to prove the effect of Mn over these special magnet microspheres, FeNa and FeMnNa catalysts were additionally prepared as comparison by one-pot method. The amounts used to synthesis all five catalysts were shown in Table 1. The molar-compositions of catalysts

were Fe100/Mnx/Na_y (x = 0, 15, y = 0, 1.6, 3.2), which were labeled as Mn15Na0(Mn15), Mn15Na16_{im}, Mn15Na32_{im}, Mn15Na32 and Mn0Na32(Na32), here the subscript 'im' indicated that Na was introduced by impregnation method while others were prepared by one-pot method, respectively.

2.2. Catalyst characterization

The surface area, pore size and pore volume of calcined samples were measured by argon physical adsorption at −186 °C on a Micromeritics ASAP 2020 instrument. The surface areas were calculated by BET method, and the pore sizes and pore volumes were measured by BJH method.

The X-ray diffraction (XRD) of the catalysts were measured on a D/MAX 2550 VB/PC (RIGAKU, Japan) with Cu Kα radiation (λ = 0.154 nm), operated at 40 Kv and 100 mA. Moreover, the scanning range was from 10° to 80° and the step was 0.02°.

The scanning electron microscopy (SEM) and high-resolution transmission electron microscopy (HRTEM) were measured on the Inspect F50 instrument and Tecnai F30 instrument at 10 kV and 300 kV, respectively. Before measurement, the catalysts were dispersed by ethanol and ultrasound to becomes more uniform. Afterwards, the energy-dispersive spectroscopy mapping (EDS-mapping) patterns were measured by a Genesis XM2 Energy Dispersive Spectrometer.

The H₂ temperature-programmed reduction (H₂-TPR) of the five samples was measured on a Micromeritics ASAP 2020 instrument. About 50 mg of catalyst (60–120 mesh) was treated in 10 % H₂/90 % Ar (v/v) at a flow rate of 50 mL/min. The measured temperature was from room temperature to 900 °C with the heating rate was 10 K/min, and the desorbed products were detected by TCD detector.

The CO temperature-programmed desorption (CO-TPD) was measured on a Micromeritics Autochem 2920 instrument. The calcined catalysts were pre-reduced in H₂ flow at 350 °C for 2 h and then cooled to room temperature under the He atmosphere. Afterwards, CO was introduced into the catalyst bed for 30 min, and then purged by He flow for 60 min. Subsequently, the sample was heated to 800 °C with the heating rate of 10 °C/min.

X-ray photoelectron spectroscopy (XPS) measurements were measured on the American Thermo ESCALAB 250Xi with Al Kα (hν = 1486.6 eV), 150 W and 500 μm beam spot.

The Mössbauer spectroscopy of catalyst samples after reaction was measured on an MR-351 constant-acceleration Mössbauer spectrometer. Measurement was conducted at room temperature with the radioactive source of ⁵⁷Co/Pd, and the spectra were collected over 512 channels in the mirror image format. Mössbauer parameters including isomer shift (IS), quadruple splitting (QS), and hyperfine field (H_{hf}) were used to identify the components of iron phase.

2.3. Fischer–Tropsch synthesis catalytic evaluation

The FTS performances of the catalysts were conducted in fixed bed reactor (10 mm i.d.). The catalyst (0.5 g) was diluted to 1:2 (mass/mass) with the same size quartz before loaded into the reactor. Then the catalysts were reduced in the atmosphere of syngas (H₂/N₂ = 1) at 350 °C and 0.1 MPa with the space velocity (GHSV) of 5000 mL/(h·g_{cat}) for 10 h. After the catalysts were cooled down to a lower temperature which was 216 °C, syngas (60 vol %H₂, 30 vol %CO, 10 vol %N₂) was introduced into the reactor with the GHSV of 11000 mL/(h·g_{cat}) to study the catalytic activity at 320 °C and 1.5 MPa.

The tail gas from the back pressure regulator was flushed into an 7890B (Agilent) gas chromatography (GC) system for on-line analysis. After the reaction was stable, data were collected and tested once half an hour. Besides, liquid phase products were analyzed off-line by a GC Agilent 7890A equipped with an Agilent 5975C mass spectrometry.

CO conversion (X_{CO}, %), CO₂ selectivity (S_{CO₂}, %) and hydrocarbon distribution (S_{C_iH_j}, %) of each hydrocarbon were calculated by formulas

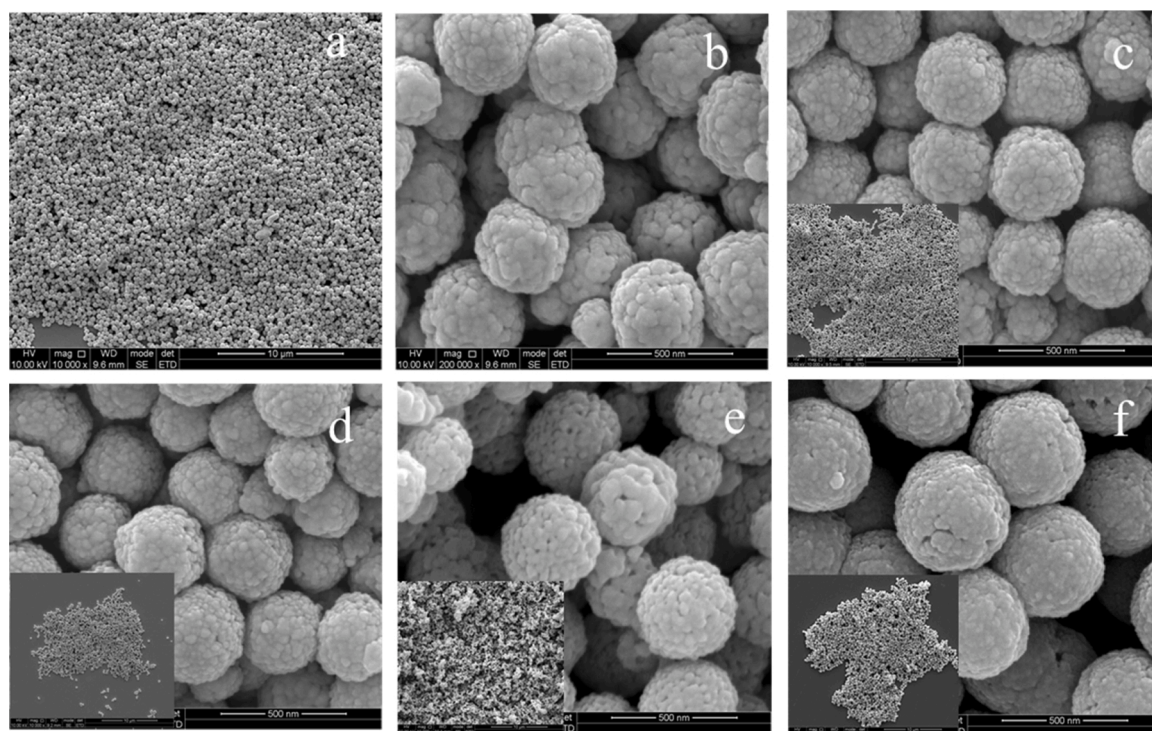


Fig. 1. SEM patterns of calcined catalysts (a, b) Mn15, (c) Mn15Na16_{im}, (d) Mn15Na32_{im}, (e) Mn15Na32, and (f) Na32.

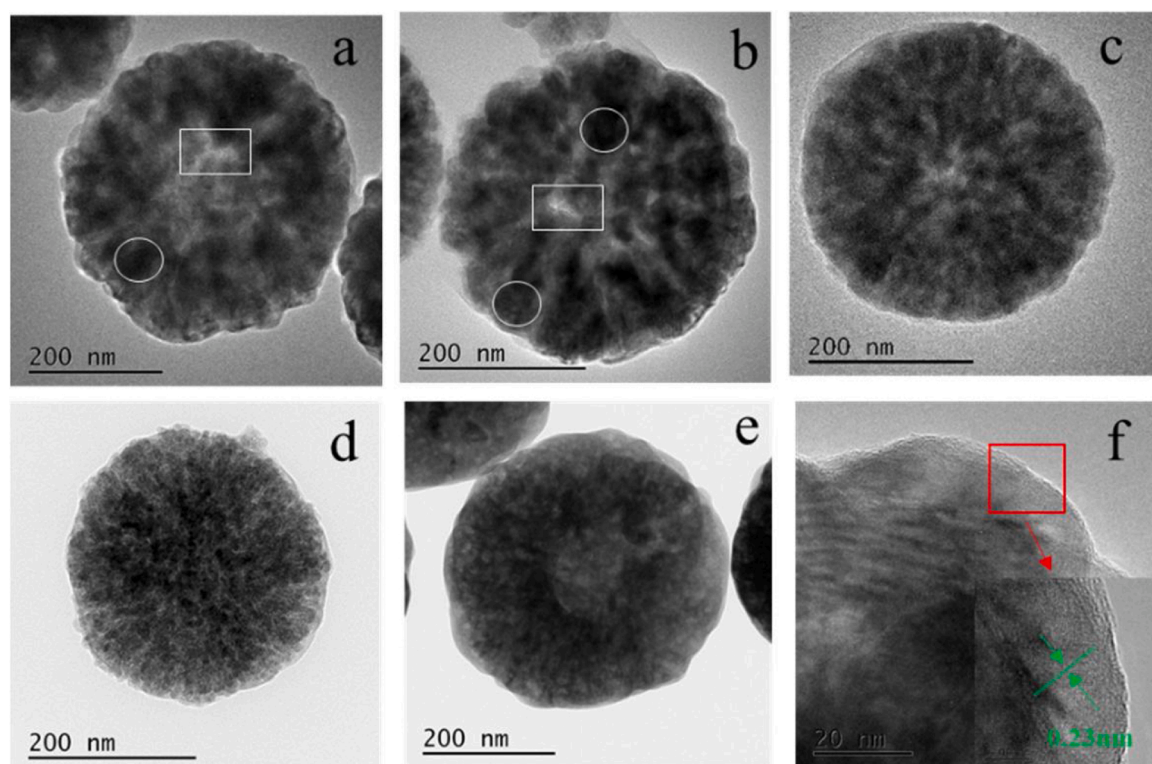


Fig. 2. HRTEM patterns of calcined catalysts (a) Mn15, (b) Mn15Na16_{im}, (c) Mn15Na32_{im}, (d) Mn15Na32, (e) Na32, and (f) the Lattice of Mn15Na32_{im}.

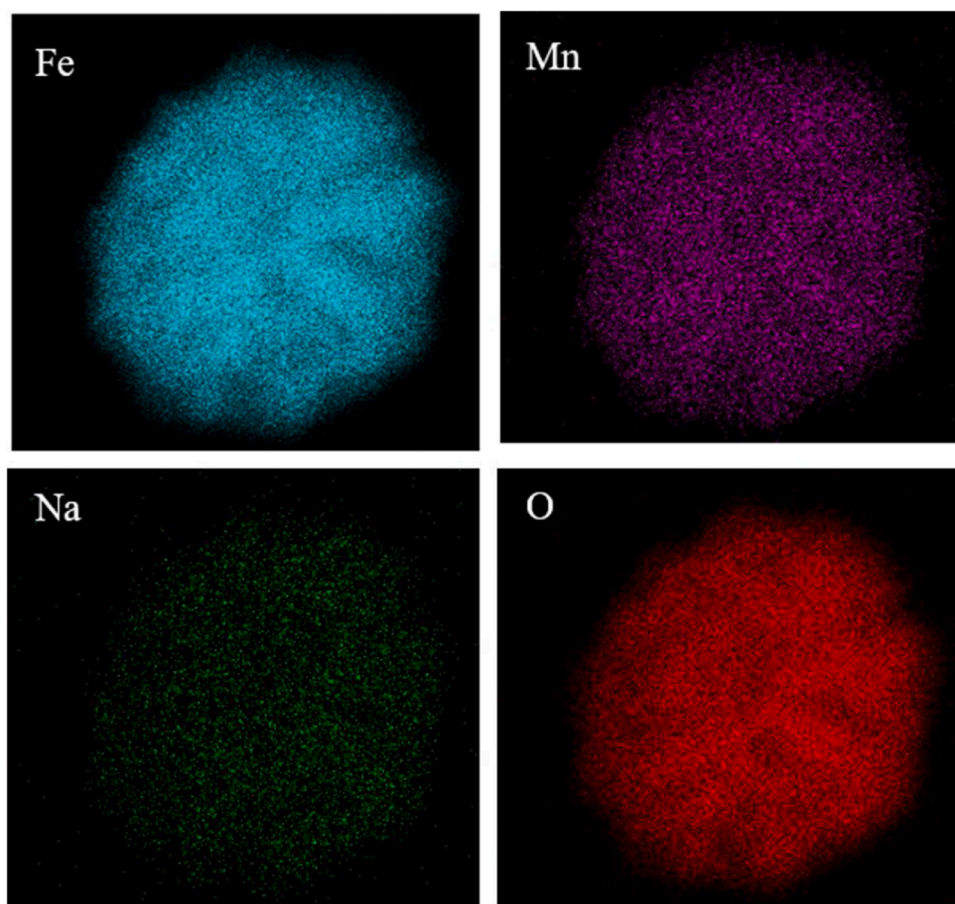


Fig. 3. EDS Mapping patterns of Mn15Na32_{im}.

1–3, respectively.

$$X_{\text{CO}}(\%) = \frac{N_{\text{in, CO}} - N_{\text{out, CO}}}{N_{\text{in, CO}}} \times 100\% \quad (1)$$

$$S_{\text{CO}_2}(\%) = \frac{N_{\text{out, CO}_2}}{N_{\text{in, CO}} - N_{\text{out, CO}}} \times 100\% \quad (2)$$

$$S_{\text{C}_i\text{H}_j}(\%) = \frac{iN_{\text{out, C}_i\text{H}_j}}{\sum_{i=1}^n iN_{\text{out, C}_i\text{H}_j}} \times 100\% \quad (3)$$

where $S_{\text{C}_i\text{H}_j}$ was based on a carbon atom basis and N_{in} and N_{out} represented molar flow rate of the inlet and outlet gas, respectively. In addition, Selectivity of Oxygenated products (S_{oxy}) was calculated based on carbon atoms in oxygenates, $N_{\text{in, CO}}$ and $N_{\text{out, CO}}$.

3. Results and discussion

3.1. Textural properties of the catalysts

Fig. 1 showed the SEM patterns of microsphere catalysts after calcination. It could be obviously seen that these magnet catalysts appeared to be a kind of highly uniform sphere, and the nanospheres were with the size of approximately 300–400 nm. Moreover, HRTEM pattern (Fig. 2) showed that larger microspheres were clustered together by many smaller nanoparticles assembled. This special topography was due to solvothermal synthesis path. First of all, Fe^{3+} was hydrolyzed to form FeOOH as an essential intermediate. Then FeOOH was partially reduced to $\text{Fe}(\text{OH})_2$ under the influence of EG, and formed Fe_3O_4 nanocrystals through dehydration. The adjacent nanocrystals assembled to magnet nanosphere by self-assembling in the last step [14]. It could be

seen in Fig. 1 that the diameter of FeNa microspheres in Fig. 1f was approximately 400 nm. While FeMn or FeMnNa microspheres owned smaller diameter of about 300 nm. The addition of Mn seeds on Fe_3O_4 corresponded with a slower aggregation rate, and thus resulted in smaller microspheres. On the other hand, some studies [15,16] had reported that the addition of Mn would cause a strong interaction between Fe and Mn then formed FeMn composite oxide ($\text{Fe}_x\text{Mn}_y\text{O}_z$). Therefore, it could demonstrate that Mn participated in the above nucleation process of ferric oxide, then led to lower crystallization and formation of amorphous species in the resultant magnetic particles [7].

HRTEM measurements were adopted to investigate the microsphere catalysts after calcination. It could be seen in Fig. 2 that metal promoted Fe_3O_4 microsphere catalysts exhibited a uniform spherical shape, and the diameter of microsphere was approximately 300–400 nm. The result of HRTEM was in good agreement with SEM results. It was further approved this magnetite particle was composed of plentiful nanocrystals with the size of about 10–20 nm. The area marked with circle in Fig. 2a and 2b corresponded to the crystalline phase while area marked with box corresponded to the amorphous phase in the particles, suggesting that small nanoparticles in large microspheres were connected with each other by the amorphous matrix [17]. Fig. 2 indicated that microsphere catalysts were in shape of loose clusters, and some areas were concentrated while some were less concentrated. The structure of magnetite particles was similar to the appearance of a ‘pomegranate’, where the seeds were tightly connected to form the whole fruit. In Fig. 2e, nanoparticles inside the Mn-free catalyst (Na32) was assembled more tightly, and lead to a denser sphere. This result further indicated Mn participated in the nucleation process of ferric oxide, and caused looser sphere and lower crystallization. In addition, the lattice fringes of Mn15Na32_{im} was shown in Fig. 2f. The distance between two adjacent

Table 2
Textural property of the calcined catalysts.

Catalysts	Surface area (m ² /g)	Pore size (nm)	Pore volume (cm ³ /g)	Crystallite size ^a (nm)
Mn15	17.3	16.6	0.084	20.6
Mn15Na16 _{im}	27.1	11.2	0.090	20.4
Mn15Na32 _{im}	11.4	16.2	0.058	18.2
Mn15Na32	14.2	13.3	0.061	26.4
Na32	9.80	10.5	0.045	33.5

^a Calculated by Scherrer's equation.

planes was measured to be 0.23 nm, corresponding to the (400) lattice planes of Fe₃O₄.

EDS-mapping was also adopted to study the distribution of elements on Fe₃O₄ microspheres. The actual distribution of Fe, O, Mn, and Na elements was shown in Fig. 3, respectively. It was obviously that Mn was quite evenly distributed on the whole catalyst. Thus, Mn was doped into Fe₃O₄ microspheres, then combined with Fe₃O₄ nanoparticles to constitute the FeMn composite microspheres. Furthermore, the Na element could also be clearly seen with a small amount of uniform spherical distribution. It suggested that Na was successfully impregnated to the surface of the Mn15 catalyst. Besides, it was further proved that the dispersibility of metal promoter on such magnetic microspheres was much better than that of traditional co-precipitated iron-based catalysts [18], which may lead to better catalytic performance.

The BET surface areas, pore size and pore volume were shown in Table 2. It could be observed in Table 2 that the surface areas of five calcined catalysts were about 10 to 30 m²/g and with rare pores (pore volumes were less than 0.1 cm³/g) which indicated that the microspheres were almost solid [6]. It further illustrated that the single ferrite was formed into a microsphere through such dense assembly process and with nonporous inside [19]. Mn15 had larger BET surface area and smaller crystallite size than Mn15Na32 and Na32 which were both prepared by one-pot method. Mn could decrease the aggregation rate of Fe₃O₄ nanocrystals which was mentioned before. Therefore, manganese participated in the nucleation process of ferric oxide and decreased the aggregation rate, lead to smaller crystallite size and larger surface area. While the introduction of Na by one-pot method would affect the interaction between Mn and Fe₃O₄ nanocrystals during above process, and resulted in smaller surface area [20]. Moreover, BET surface area was increased with the lower addition of Na on Mn15 (Mn15Na16_{im}) while it was decreased with the excessive addition of Na (Mn15Na32). It suggested excessive sodium would block part of surface of Fe₃O₄ microspheres.

Fig. 4 showed the XRD pattern of the five calcined magnetic

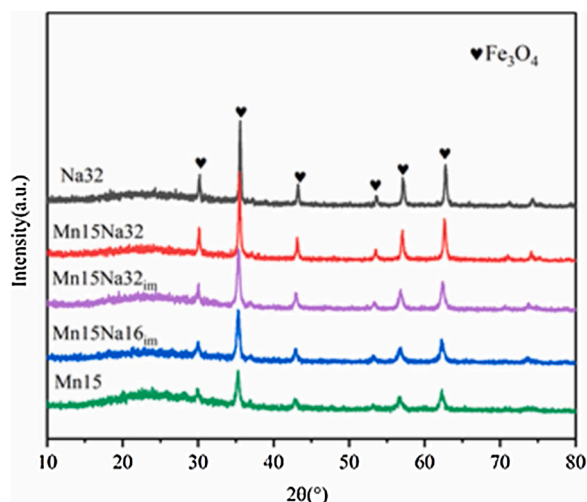


Fig. 4. XRD patterns of the calcined catalysts.

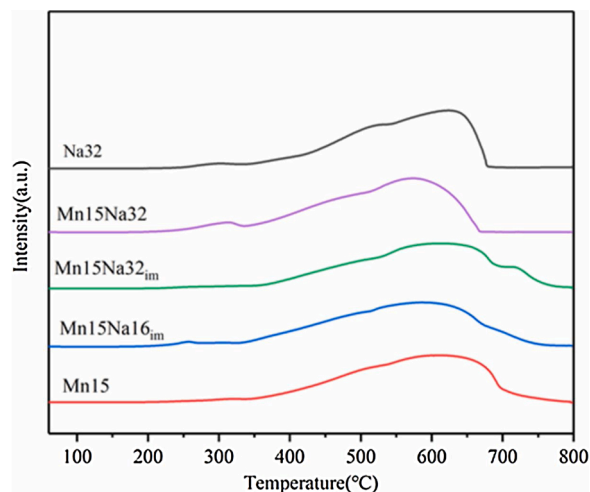


Fig. 5. H₂-TPR profiles of the calcined catalysts.

microsphere catalysts with Mn or Na promoter. It was obviously seen in Fig. 4 that the iron phase of microsphere catalysts was the well crystallized Fe₃O₄, which had characteristic peaks at 2θ values of 30.1°, 35.4°, 43°, 53.6°, 57.1° and 62.6° (Fe₃O₄, JCPDS 19-0629). It could be seen that there were no characteristic peaks shown of both Mn and Na, which mainly because Mn and Na owned good dispersibility on magnet microspheres [21]. The XRD diffraction peak intensity of Mn15 was significantly weaker than Na32 catalyst. It was suggested in previous studies [22,23] that the introduction of the Mn promoter on iron-based catalysts could stabilize the iron oxide phases by forming FeMn spinel then enhance the crystallinity. While in this study, Mn-promoted Fe₃O₄ catalysts showed both the weaker diffraction peak intensity in XRD pattern and the lower crystallinity in SEM pattern. These results could be attributed to the nucleation process of Fe₃O₄ microspheres prepared by one-pot solvothermal method, leading to different interaction between Fe and Mn from iron-based catalysts prepared by coprecipitation. Besides, with further addition of sodium on Mn15, the intensity of diffraction peaks had a visible increasing trend. It indicated that the addition of Na enhanced the Fe₃O₄ crystallinity.

3.2. Reducibility and CO chemisorption of the catalysts

Fig. 5 presented the H₂-TPR profiles of five calcined microsphere catalysts. It could be seen in Fig. 5 that there were two main reduction peaks of five calcined catalysts. The small peaks located at about 300 °C

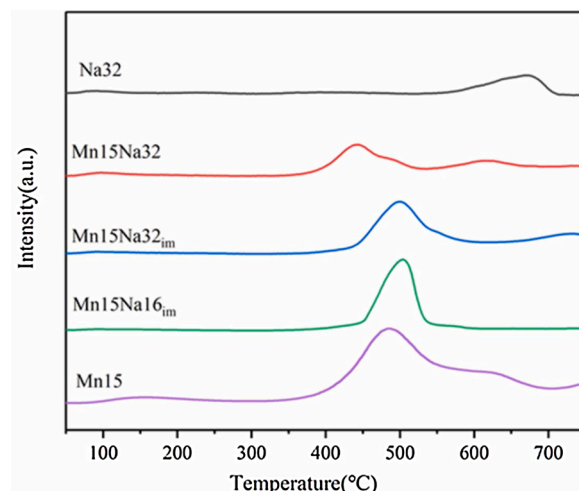


Fig. 6. CO-TPD profiles of the calcined catalysts.

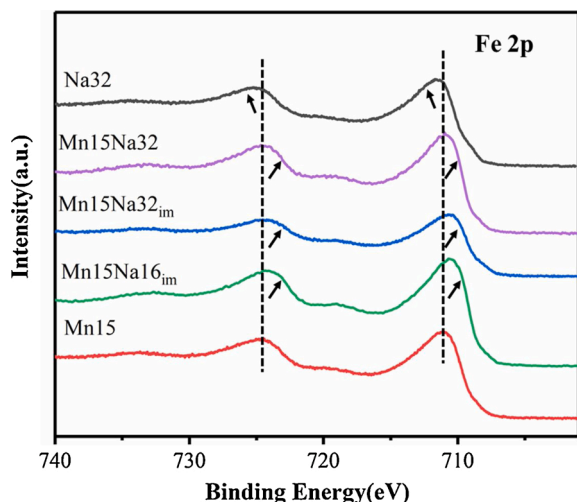


Fig. 7. XPS spectra of Fe 2p of the calcined catalysts.

could be attributed to Fe_2O_3 which oxidized by Fe_3O_4 on the surface of microsphere catalysts [24]. The peaks located in the 550–650 °C range could be assigned to the continual reduction of Fe_3O_4 to FeO and FeO to $\alpha\text{-Fe}$, respectively [25]. The H_2 -TPR results also revealed that Mn partly suppressed the reduction of the Fe_3O_4 microspheres catalysts. The peak of Mn15 catalyst was at the highest temperature among the 5 catalysts which indicated that Mn15 owned worse reducibility in H_2 . With the introduction of Na by impregnation on the surface of Mn15, the peaks of Mn15Na16_{im} and Mn15Na32_{im} shifted to lower temperature slightly. While it could be obviously seen that the peaks of Mn15Na32 and Na32 shifted to much lower temperature area. These results could be attributed to the strong interaction between Fe and Mn which was proved by SEM, Ar-physisorption and XRD. The Fe-Mn interaction was attributed to the entry of Mn ions into the Fe lattice, and it could inhibit catalyst reduction [13]. The presence of manganese on Fe_3O_4 microspheres would reduce the hydrogen chemisorption, resulting in a reduction inhibition of the manganese promoted catalyst in H_2 [22]. In addition, Na as an alkali metal promoted the reduction of Fe-Mn catalyst in H_2 atmosphere due to its basicity and electronic donation [26]. While comparing Mn15Na16_{im} and Mn15Na32_{im}, catalyst loaded with more Na on the surface moved to a higher reduction temperature. It was considered that Na located on the surface of Fe_3O_4 , leading to blocking part of surface Fe_3O_4 and further enhancing the Fe-Mn interaction [24].

CO-TPD was used to investigate the chemisorption properties of five calcined catalysts, and the results were shown in Fig. 6. Al-Dossary et al. [27] reported that traditional iron-based catalysts had two groups of CO desorption peaks. For the series of Fe_3O_4 catalysts in this study, it could be seen in Fig. 6 that the weak CO desorption peaks at about 100–150 °C were apparent to be such weak while the strong CO desorption peaks at 450–550 °C were visible. Thus, the type of CO adsorption on these magnet catalysts was mainly strong adsorption. It could be obviously seen that Mn15, Mn15Na16_{im} and Mn15Na32_{im} showed relatively stronger CO desorption peaks which indicated that the CO adsorption was stronger on these 3 catalysts. This result may be due to the influence of Mn over the spherical structure and CO adsorbed state of Fe_3O_4 since both Mn15Na16_{im} and Mn15Na32_{im} were prepared based on Mn15. It was proved by SEM and XRD that Mn lead to lower crystallinity of Fe_3O_4 and could inhibit the stability of iron oxide phase. Feyzi et al. [28] suggested that the stable iron phase would suppress the CO adsorption and catalyst carburization. Thus, the lower crystallinity of Mn-promoted Fe_3O_4 was beneficial for strong CO adsorption in this study. In contrast, the CO desorption of Na32 shifted to higher temperature while the peak intensity was pretty low which may be caused by the uniform spherical structure and higher crystallinity of Na32.

On the other hand, the surface-enriched Mn on magnet microspheres

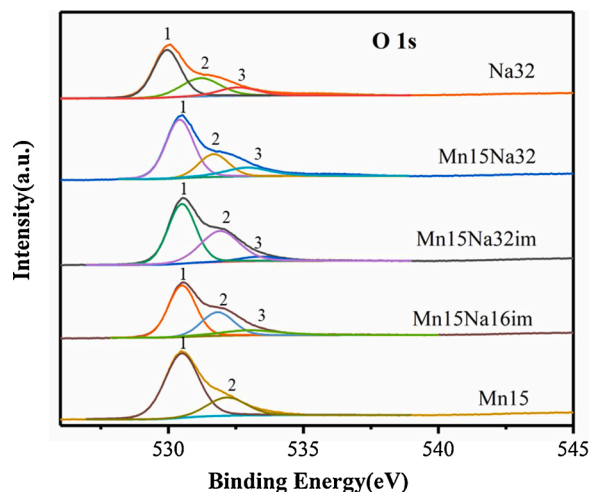


Fig. 8. XPS spectra of O 1s of the calcined catalysts.

according to the EDS-mapping results would provide CO dissociation site which may benefit the Fe-C bond [29] and cause a higher CO concentration over the surface of microsphere catalysts, thus enhancing the CO adsorption [30,31]. Considering the effect of Na introduced on the Mn15 catalyst by impregnation, the variety of CO desorption peaks was not significant, indicating that the effect of sodium on CO adsorption was not significant. However, sodium introduced by one-pot method on Fe_3O_4 catalysts showed lower intensity of CO desorption peaks where the CO adsorption of Mn15Na32 was suppressed compared to Mn15Na32_{im}. It was considered that Na introduced by one-pot method caused an impact on the Fe-Mn interaction during the nucleation process of Fe_3O_4 , then suppressed the CO adsorption of catalyst.

3.3. Electronic effect of the catalysts

The XPS spectra of the five calcined catalysts were shown in Figs. 7 and 8. The Fe 2p_{3/2} and Fe 2p_{1/2} peaks centered at 711 and 725 eV were in good agreement with the standard Fe_3O_4 phase [31], as shown in Fig. 7. This result was consistent well with the XRD results in Fig. 4. It could be seen that compared to Mn-free catalyst (Na32), the peaks of FeMn and FeMnNa microsphere catalysts shifted toward lower binding energies. This result indicated the variation of the electron density around the atom. It is known that the electron binding energy is lower with the greater electron density [32]. Therefore, Mn could be regarded as electronic donating promoter, then transferred electrons to Fe element and led to an electron-rich state of Fe species. Meanwhile, the electrons donation of Mn could facilitate the desorption of CO then cause a high CO concentration over the surface of microsphere catalysts, and it was consistent well with the CO-TPD results. It also proved that Mn didn't only change the spherical structure of microspheres, but also changed the electronic state of Fe. Besides, the peaks of Mn15Na16_{im}, Mn15Na32_{im} and Mn15Na32 catalysts shifted toward lower binding energies compared to Mn15. It further demonstrated that Na could also donate electronics to Fe-Mn catalyst then improved the FTS reaction activity.

As shown in Fig. 8, three O 1s peaks centered at about 530, 531.2 and 532.6 eV could be observed of FeNa and FeMnNa catalysts. These O 1s peaks were belonged to O^{2-} from Fe_3O_4 , and were divided according to the different ion states [33]. While only two O 1s peaks centered at 530.5 and 532.3 eV could be observed for Mn15. Therefore, the consequence that Mn-modified Fe_3O_4 catalyst only showed 2 peaks of O 1s which was considered that Mn^{2+} may occupy the tetrahedral site of Fe^{3+} then occupied oxygen vacancies during the process of nucleation in one-pot method, and lead to the decreasing number of O1s peaks of O^{2-} from Fe_3O_4 [34]. What's more, the main peak of O1s of FeMn and

Table 3
MES parameters of the catalysts after reaction.^a

Catalysts	Phases	Mössbauer parameters			
		IS (mm/s)	QS (mm/s)	Hhf (kOe)	Area (%)
Mn15	Fe ₃ O ₄	0.32	−0.02	489	13.7
		0.61	0.01	453	24.9
		0.27	−0.1	214	9.7
	χ-Fe ₅ C ₂	0.23	−0.08	171	38.2
		0.6	0.59	94	3.8
Mn15Na32 _{im}	Fe ³⁺ (spm)	0.31	1.02		9.7
	Fe ₃ O ₄	0.37	0.02	489	3.4
		0.76	0.25	456	3.7
	χ-Fe ₅ C ₂	0.26	−0.08	217	29.4
		0.22	0	178	41.8
Na32	Fe ₃ O ₄	0.14	−0.04	101	21.7
		0.31	−0.01	491	6.6
		0.65	0.02	458	12.6
	χ-Fe ₅ C ₂	0.26	−0.09	216	30.5
		0.2	0	184	32.7
		0.2	−0.05	109	17.6

^a Reaction conditions: 320 °C, 1.5 MPa, H₂/CO = 2, 11000 mL/(h·g_{cat}), 24 h.

FeMnNa catalysts shifted to higher binding energy than FeNa catalyst (Na32). O element of these 4 catalysts owned higher binding energy and lower electron density of O^{2−}. It was considered that Na owned the ability to donate electrons as the alkaline promoter. Hence sodium was introduced on Mn-promoted Fe₃O₄ catalysts to further enhance the CO dissociative adsorption, thereby improving the FTS activity.

3.4. Mössbauer emission spectroscopy

The MES parameters of the samples after reaction were shown in Table 3. As shown in Table 3, the iron-based catalysts after reaction mainly consisted of Fe₃O₄, χ-Fe₅C₂ and Fe³⁺. The sextets with Hhf of about 489–491 and 453–458 kOe could be attributed to Fe₃O₄ (A and B sites), respectively. The fitted Hhf value of 214–217, 171–184 and 94–109 kOe consisted with that of χ-Fe₅C₂ [35]. In addition, the IS values of the central doublet could be attributed to Fe³⁺ ion. The Fe³⁺ ion was in the state of octahedral superparamagnetic with small particles. χ-Fe₅C₂ phase was known to be the primary active phase indicators for FTS reaction [36]. It could be seen that compared to Na32 and Mn15, Mn15Na32_{im} was composed of the highest amount of χ-Fe₅C₂ (92.9 %) and the lowest amount of Fe₃O₄ (7.1 %). For traditional Fe₂O₃ catalysts, the transformation process of iron species was that iron oxide was firstly reduced from Fe₂O₃ to Fe₃O₄, and then transformed to FeC_x in FTS. While as for magnet microsphere catalysts in this study, Fe₃O₄ was directly transformed to FeC_x for FTS [22]. Therefore, higher amount of χ-Fe₅C₂ for Fe-Mn-Na catalyst than Fe-Mn and Fe-Na catalysts indicated that both Mn and Na could promote the transformation of Fe₃O₄ to χ-Fe₅C₂ in FTS. It's worth noting that with the addition of Na, the content of χ-Fe₅C₂ increased from 51.7%–92.9% which was a significant improvement. The content of χ-Fe₅C₂ increased from 80.8% to 92.9% with the effect of Mn promoter. Yet it was also suggested by Wu et al. [37] that manganese could not only improve the amount of χ-Fe₅C₂, but

also change the species of iron carbides then promote syngas to light olefins. Both Mn and Na promoters could enhance the carbon deposition and transformation to χ-Fe₅C₂. It was consistent well with the CO-TPD and XPS results. The high content of χ-Fe₅C₂ also lead to a high activity of FTS reaction in Table 3. In addition, it couldn't be observed the presence of Fe³⁺ (spm) in MES results of Na32 and Mn15Na32_{im}. It was considered that sodium was helpful for the iron oxide activation as the alkali promoter.

3.5. FTS reaction performance of catalysts

The FTS activity and product selectivity of microsphere catalysts were shown in Table 4. The oxygenates in Table 4 were mainly ethers and lower alcohol such as ether and ethanol. It was obviously showed in Table 4 that Fe₃O₄ catalyst with only manganese promoted (Mn15) exhibited the lowest CO conversion which of 77.47 %. With the further introduction of sodium, the CO conversion increased and reached the maximum value of 97.21 % on Mn15Na32_{im}. It is known that χ-Fe₅C₂ is considered to be the primary active phase indicators for FTS reaction [36]. Hence the highest content of χ-Fe₅C₂ of Mn15Na32_{im} was consistent with the highest CO conversion. Na could donate electronics to Fe species, enhanced the catalysts reduction in H₂ and facilitated the χ-Fe₅C₂ formation which was proved by XPS, H₂-TPR and MES. As a result, Na improved the activity of FTS reaction and lead to a higher CO conversion. Interestingly, comparing with Mn15Na32 and Mn15Na32_{im}, catalyst prepared by one-pot method exhibited lower CO conversion of 88.3 % than impregnation method. It was mentioned before that the Fe-Mn interaction emerged with the addition of Mn on magnet microspheres. Mn could donate electronics to Fe₃O₄ and promote CO dissociation adsorption with the above interaction. Na introduced by impregnation based on Fe-Mn microsphere would not affect this interaction. While Na introduced by one-pot method may alter the effect of Mn on the electronic state of Fe₃O₄, then resulted in lower CO conversion.

The CO₂ selectivity for all the catalysts was shown in Table 4. It could be seen that the CO₂ selectivity of Mn15 was the lowest which was 28.91 %. CO₂ selectivity tended to increase slowly with the introduction of sodium. Fe₃O₄ was normally considered as an active phase of water-gas shift reaction. While Both Liu et al. and Wu et al. [37,38] suggested that Fe₃O₄ could suppressed the formation of CO₂. In contrast, χ-Fe₅C₂ played a key role in promoting the Boudouard mechanism (2CO → C + CO₂), and lead to higher CO₂ selectivity. Comparing Mn15, Mn15Na32_{im} and Na32 in this study, Mn15 exhibited the highest Fe₃O₄ content and lowest χ-Fe₅C₂ content in MES results. Thus, the lowest CO₂ selectivity of Mn15 was consistent well with the above suggestion that χ-Fe₅C₂ facilitated the formation of CO₂.

The hydrocarbon distribution was also shown in Table 4. A series of microsphere catalysts showed the selectivity of light olefins (C₂–C₄) in the range of 29.51%–38.36%. It was reported that Fe₃O₄ microspheres effectively reduced the second reaction of formed olefins, leading to higher olefins selectivity [39]. Therefore, all 5 microsphere catalysts in this study showed relatively higher light olefins selectivity. The Na32 catalyst exhibited the lowest C₂–C₄ selectivity of 29.51 % while

Table 4
CO conversions and hydrocarbons selectivity of different catalysts.^a

Catalyst	X _{CO} (%)	S _{CO2} (%)	S _{Oxy} (%)	Distribution of hydrocarbons (%)				O/P (C ₂ -C ₄)	Carbon Balance (%)
				CH ₄	C ₂ –C ₄	C ₂ °–C ₄ °	C ₅ +		
Mn15	77.47	28.91	1.86	16.98	30.89	11.83	40.30	2.61	98.0
Mn15Na16 _{im}	96.47	30.34	2.91	10.39	30.18	4.53	54.90	6.66	97.4
Mn15Na32 _{im}	97.21	31.91	2.45	12.41	33.02	6.20	48.38	5.33	97.7
Mn15Na32	88.30	32.59	3.23	19.65	38.36	9.96	32.03	3.85	96.5
Na32	97.03	34.05	3.09	23.66	29.51	15.32	31.52	1.93	97.2

^a Reaction conditions: 320 °C, 1.5 MPa, H₂/CO = 3, 11000 mL/(h·g_{cat}), 24 h.

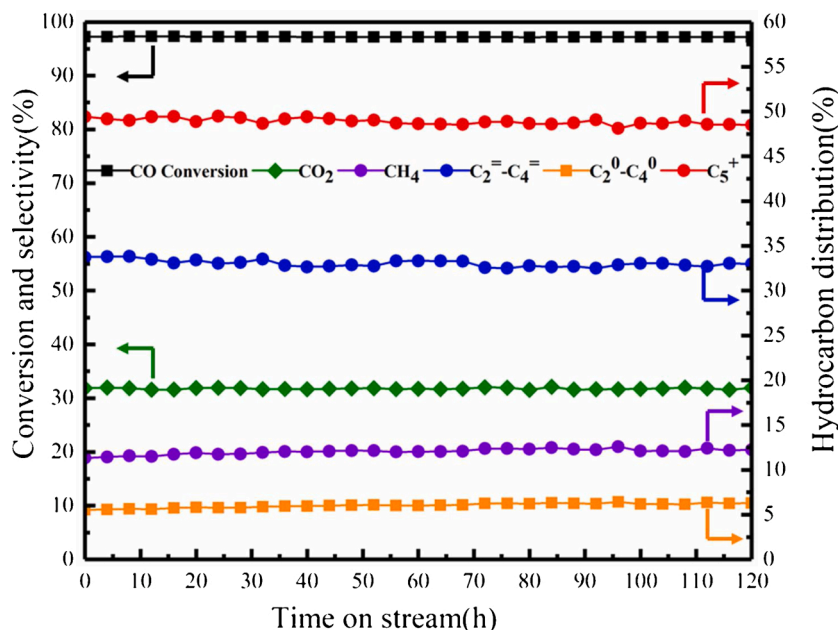


Fig. 9. Stability test of the Mn15Na32_{im} catalyst. Reaction conditions: 320 °C, 1.5 MPa and 11000 mL/(h·g_{Cat}).

Mn15Na32 showed the highest C₂–C₄ selectivity of 38.36 %. This result was in compliance with those of H₂-TPR and CO-TPD. Mn promoted the chemisorption of CO, and prevented olefins from secondary hydrogenation. The Fe-Mn interaction proved by XRD and XPS improved the inter-dispersion effect of manganese on iron phases, and it weakened the connection of initial formed olefin, thereby reduced the secondary hydrogenation of the primary product [22]. Compared to Na32, Mn15Na32 showed lower selectivity of C₂–C₄ which was 9.96 % and higher ratio of C₂–C₄ olefin/paraffin (O/P) which was 3.85. The higher value of O/P further indicated the suppression of the hydrogenation reaction. On the other hand, the C₂–C₄ selectivity of Fe₃O₄ catalyst with only manganese promoted (Mn15) was 30.89 %. With the impregnation of Na on Mn15 catalyst, there was no obvious variety in C₂–C₄ selectivity with a lower amount of Na (Mn15Na16_{im}). C₂–C₄ selectivity increased to 33.02 % with a larger amount of Na (Mn15Na32_{im}). Alkali metal was considered to enhance the surface basicity [12]. Therefore, sodium promoter could result in a high concentration ratio of CO to H₂ on catalyst surface. This result would lead to a high selectivity of light olefins and low hydrogenation activity. Besides, Mn15Na32_{im} had higher content of χ -Fe₅C₂ after reaction than Mn15 which also exhibited higher selectivity of C₂–C₄. It illustrated that χ -Fe₅C₂ may not only enhance the FTS reaction activity, but also promote syngas to light olefins. In addition, the stability test was conducted over Mn15Na32_{im} catalyst for 120 h, and the result was shown in Fig. 9. It could be obviously seen in Fig. 9 that the CO conversion, CO₂ selectivity and hydrocarbon selectivity kept basically stable during the 120 h-reaction. The result of stability test indicated that metal-promoted microsphere catalyst exhibited superior stability.

4. Conclusion

Mn-promoted Fe₃O₄ catalysts were prepared by one-pot solvothermal method then Na was further introduced by one-pot method or impregnation to improve the activity and C₂–C₄ selectivity of FTS reaction. Mn could affect the electronic state of magnet microspheres via the Fe-Mn interaction, and increased the BET surface area. CO-TPD and H₂-TPR indicated that Mn promoted the dissociation and adsorption of CO, suppressed the reduction of catalyst in H₂. The addition of Na enhanced the FTS reaction activity by its ability to donate electronic and enhance the surface basicity of catalysts. The MES results proved that

both Mn and Na promoted the conversion of Fe₃O₄ to χ -Fe₅C₂ in FTS. The Mn15Na32_{im} catalyst prepared by impregnation exhibited the highest CO conversion of 97.21 % with the C₂–C₄ selectivity of 33.02 %. Mn15Na32_{im} also showed the O/P value of 5.33, indicating that Na impregnated on the surface of catalysts may suppress the hydrogenation reaction and lead to higher value of O/P. In addition, the Mn15Na32 catalyst displayed the best C₂–C₄ selectivity of 38.36 % compared to the Na32 catalyst which demonstrated that Mn promoted the chemisorption of CO, while suppressed the secondary hydrogenation of olefins, then lead to higher selectivity of light olefins.

CRediT authorship contribution statement

Hewei Zhang: Conceptualization, Methodology, Investigation, Writing - original draft, Visualization. **Haitao Zhang:** Resources, Funding acquisition. **Weixin Qian:** Funding acquisition. **Xian Wu:** Writing - review & editing. **Hongfang Ma:** Writing - review & editing, Supervision, Funding acquisition. **Qiwen Sun:** Funding acquisition. **Weiyong Ying:** Writing - review & editing, Supervision, Funding acquisition.

Declaration of Competing Interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript entitled “Sodium Modified Fe-Mn Microsphere Catalyst for Fischer–Tropsch Synthesis of Light Olefins”.

Acknowledgements

We gratefully acknowledge the financial supports from the National Natural Science Foundation of China (grant no. 21706068), the National High Technology Research and Development Plan of China (863 plan, 2011AA05A204), and the Fundamental Research Funds for the Central Universities (no. 50321012017013).

References

- [1] J. Biswas, I.E. Maxwell, Recent process-and catalyst-related developments in fluid catalytic cracking, *Appl. Catal.* 63 (1990) 197–258.
- [2] E. Iglesia, Design, synthesis, and use of cobalt-based Fischer-Tropsch synthesis catalysts, *Appl. Catal. A* 161 (1997) 59–78.
- [3] L. Zhong, F. Yu, Y. An, Y. Zhao, Y. Sun, Z. Li, T. Lin, Y. Lin, X. Qi, Y. Dai, L. Gu, J. Hu, S. Jin, Q. Shen, H. Wang, Cobalt carbide nanoprisms for direct production of lower olefins from syngas, *Nature* 538 (2016) 84–87.
- [4] S. Abello, D. Montane, Exploring iron-based multifunctional catalysts for Fischer-Tropsch synthesis: a review, *ChemSusChem* 4 (2011) 1538–1556.
- [5] H. Galvis, J. Bitter, C. Khare, M. Ruitenbeek, A. Dugulan, K. Jong, Supported iron nanoparticles as catalysts for sustainable production of lower olefins, *Science* 335 (2012) 835–838.
- [6] Y. Deng, Y. Cai, Z. Sun, J. Liu, C. Liu, J. Wei, W. Li, C. Liu, Y. Wang, D. Zhao, Multifunctional mesoporous composite microspheres with well-designed nanostructure: a highly integrated catalyst system, *J. Am. Chem. Soc.* 132 (2010) 8466–8473.
- [7] J. Liu, Z. Sun, Y. Deng, Y. Zou, C. Li, X. Guo, L. Xiong, Y. Gao, F. Li, D. Zhao, Highly water-dispersible biocompatible magnetite particles with low cytotoxicity stabilized by citrate groups, *Angew. Chem. Int. Ed. Engl.* 48 (2009) 5875–5879.
- [8] C. Wang, Q. Wang, X. Sun, L. Xu, CO hydrogenation to light alkenes over Mn/Fe catalysts prepared by coprecipitation and sol-gel methods, *Catal. Lett.* 105 (2005) 93–101.
- [9] F.G. Botes, J.W. Niemantsverdriet, J. van de Loosdrecht, A comparison of cobalt and iron based slurry phase Fischer-Tropsch synthesis, *Catal. Today* 215 (2013) 112–120.
- [10] Z. Yang, X. Pan, J. Wang, X. Bao, FeN particles confined inside CNT for light olefin synthesis from syngas: effects of Mn and K additives, *Catal. Today* 186 (2012) 121–127.
- [11] Y. Xu, X. Jia, X. Liu, Supported Fe/MnOx catalyst with Ag doping for remarkably enhanced catalytic activity in Fischer-Tropsch synthesis, *Catal. Sci. Technol.* 8 (2018) 1953–1970.
- [12] H.M. Torres Galvis, A.C.J. Koeken, J.H. Bitter, T. Davidian, M. Ruitenbeek, A. I. Dugulan, K.P. de Jong, Effects of sodium and sulfur on catalytic performance of supported iron catalysts for the Fischer-Tropsch synthesis of lower olefins, *J. Catal.* 303 (2013) 22–30.
- [13] M.C. Ribeiro, G. Jacobs, R. Pendyala, B.H. Davis, D.C. Cronauer, A.J. Kropf, C. L. Marshall, Fischer-Tropsch synthesis: influence of Mn on the carburization rates and activities of Fe-based catalysts by TPR-EXAFS/XANES and catalyst testing, *J. Phys. Chem. C* 115 (2011) 4783–4792.
- [14] B. Jia, L. Gao, Morphological transformation of Fe₃O₄ spherical aggregates from solid to hollow and their self-assembly under an external magnetic field, *J. Phys. Chem. C* 112 (2018) 666–671.
- [15] H. Wang, Y. Yang, J. Xu, H. Wang, M. Ding, Y. Li, Study of bimetallic interactions and promoter effects of FeZn, FeMn and FeCr Fischer-Tropsch synthesis catalysts, *J. Mol. Catal. A Chem.* 326 (2010) 29–40.
- [16] J. Li, L. Wu, S. Zhang, J. Wen, M. Liu, C. Wang, X. Li, The promotional effect of Mn on Fe-based Fischer-Tropsch catalysts for the synthesis of C₅₊ hydrocarbons, *Sustain. Energ. Fuels* 3 (2019) 219–226.
- [17] H. Deng, X. Li, Q. Peng, X. Wang, J. Chen, Y. Li, Monodisperse magnetic single-crystal ferrite microspheres, *Angew. Chem. Int. Ed. Engl.* 44 (2005) 2782–2785.
- [18] Y. Yang, Effect of potassium promoter on precipitated iron-manganese catalyst for Fischer-Tropsch synthesis, *Appl. Catal. A Gen.* 266 (2004) 181–194.
- [19] X. Lv, J. Chen, Y. Tan, Y. Zhang, A highly dispersed nickel supported catalyst for dry reforming of methane, *Catal. Commun.* 20 (2012) 6–11.
- [20] Z. Tao, Y. Yang, H. Wan, T. Li, X. An, H. Xiang, Y. Li, Effect of manganese on a potassium-promoted iron-based Fischer-Tropsch synthesis catalyst, *Catal. Lett.* 114 (2007) 161–168.
- [21] D. Wang, X. Zhou, J. Ji, X. Duan, G. Qian, X. Zhou, D. Chen, W. Yuan, Modified carbon nanotubes by KMnO₄ supported iron Fischer-Tropsch catalyst for the direct conversion of syngas to lower olefins, *J. Mater. Chem. A* 3 (2015) 4560–4567.
- [22] T. Li, H. Wang, Y. Yang, H. Xiang, Y. Li, Effect of manganese on the catalytic performance of an iron-manganese bimetallic catalyst for light olefin synthesis, *J. Energy Chem.* 22 (2013) 624–632.
- [23] J. Zhang, L. Ma, S. Fan, T. Zhao, Y. Sun, Synthesis of light olefins from CO hydrogenation over Fe-Mn catalysts: effect of carburization pretreatment, *Fuel* 109 (2013) 116–123.
- [24] K. Keyvanloo, J.B. Horton, W.C. Hecker, M.D. Argyle, Effects of preparation variables on an alumina-supported FeCuK Fischer-Tropsch catalyst, *Catal. Sci. Technol.* 4 (2014) 4289–4300.
- [25] F. Jiang, M. Zhang, B. Liu, Y. Xu, X. Liu, Insights into the influence of support and potassium or sulfur promoter on iron-based Fischer-Tropsch synthesis: understanding the control of catalytic activity, selectivity to lower olefins, and catalyst deactivation, *Catal. Sci. Technol.* 7 (2017) 1245–1265.
- [26] J. Li, H. Ma, H. Zhang, Q. Sun, W. Ying, D. Fang, Sodium promoter on iron-based catalyst for direct catalytic synthesis of light alkenes from syngas, *Fuel Process. Technol.* 125 (2014) 119–124.
- [27] M. Al Dossary, J.L.G. Fierro, J.J. Spivey, Cu-Promoted Fe₂O₃/MgO-based Fischer-Tropsch catalysts of biomass-derived syngas, *Ind. Eng. Chem. Res.* 54 (2015) 911–921.
- [28] M. Feyzi, M. Irandoust, A.A. Mirzaei, Effects of promoters and calcination conditions on the catalytic performance of iron-manganese catalysts for Fischer-Tropsch synthesis, *Fuel Process. Technol.* 92 (2011) 1136–1143.
- [29] Y. Zhang, L. Ma, T. Wang, X. Li, MnO₂ coated Fe₂O₃ spindles designed for production of C₅₊ hydrocarbons in Fischer-Tropsch synthesis, *Fuel* 177 (2016) 197–205.
- [30] M. Ding, Y. Yang, B. Wu, J. Xu, C. Zhang, H. Xiang, Y. Li, Study of phase transformation and catalytic performance on precipitated iron-based catalyst for Fischer-Tropsch synthesis, *J. Mol. Catal. A Chem.* 303 (2009) 65–71.
- [31] Q. Guo, J. Huang, W. Qian, H. Zhang, H. Ma, W. Ying, Effect of lanthanum on Zr-Co/-γ-Al₂O₃ catalysts for Fischer-Tropsch synthesis, *Catal. Lett.* 148 (2018) 2789–2798.
- [32] S. Qin, C. Zhang, J. Xu, Y. Yang, H. Xiang, Y. Li, Fe-Mo interactions and their influence on Fischer-Tropsch synthesis performance, *Appl. Catal. A Gen.* 392 (2011) 118–126.
- [33] T. Yamashita, P. Hayes, Analysis of XPS spectra of Fe²⁺ and Fe³⁺ ions in oxide materials, *Appl. Surf. Sci.* 254 (2008) 2441–2449.
- [34] Y. Xu, J. Liu, J. Wang, G. Ma, J. Lin, Y. Yang, Y. Li, C. Zhang, M. Ding, Selective conversion of syngas to aromatics over Fe₃O₄@MnO₂ and hollow HZSM-5 bifunctional catalysts, *ACS Catal.* 9 (2019) 5147–5156.
- [35] M. Ding, Y. Yang, B. Wu, T. Wang, L. Ma, H. Xiang, Y. Li, Transformation of carbonaceous species and its influence on catalytic performance for iron-based Fischer-Tropsch synthesis catalyst, *J. Mol. Catal. A Chem.* 351 (2011) 165–173.
- [36] E. De Smit, F. Cinquini, A.M. Beale, O.V. Safonova, W. Beek, P. Sautet, B. M. Weckhuysen, Stability and reactivity of ε-γ-iron carbide catalyst phases in Fischer-Tropsch Synthesis: controlling μc, *J. Am. Chem. Soc.* 132 (2010) 14928–14941.
- [37] X. Wu, H. Ma, H. Zhang, W. Qian, D. Liu, Q. Sun, W. Ying, High-temperature Fischer-Tropsch synthesis of light olefins over nano-Fe₃O₄@MnO₂ core-shell catalysts, *Ind. Eng. Chem. Res.* 58 (2019) 21350–21362.
- [38] Y. Liu, J. Chen, J. Bao, Y. Zhang, Manganese-modified Fe₃O₄ microsphere catalyst with effective active phase of forming light olefins from syngas, *ACS Catal.* 5 (2015) 3905–3909.
- [39] H. Schulz, Selforganization in Fischer-Tropsch synthesis with iron- and cobalt catalysts, *Catal. Today* 228 (2014) 113–122.